```
(*
                              CS51 Lab 17
                           Objects and Classes
 *)
(*
                                SOLUTION
 *)

(*======================================================================
Readings:

    This lab builds on material from Chapter 18 of the textbook
    <http://book.cs51.io>, which should be read before the lab session.

Objective:

    This lab provides practice with object-oriented programming: the
    creation of classes, interfaces, inheritance, subtyping, and dynamic
    dispatch.
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Part 1: Flatland

Suppose we want to model a two-dimensional world called Flatland
populated by creatures of geometric shapes
(https://url.cs51.io/flatland).  In particular, we want:

  o All shapes to have some notion of *area*, so that when we meet any
    new shape, we can easily calculate the shape's area.

  o To know the shape's *location* in Flatland space.

  o To support many different kinds of shapes.

How can we solve this problem?

There are several approaches. As a first attempt we might use
algebraic data types. (Later, we'll see this isn't quite ideal.)

First, we'll define a point to store (x, y) coordinates. *)

type point = float * float ;;

(* Now we'll define `shape_adt`, an algebraic data type for shapes,
with the ability to represent some shapes: a `Square`, a `Rect`, and a
`Circle`. Each shape will have some aspects that together specify its
location and size, as follows:

  Square -- a single point for the location of its lower-left corner
    and an edge length.

  Rect -- a single point for the location of its lower-left corner, a
```

```
    width, and a height.

  Circle -- a single point for the location of its center and a
    radius.
 *)

type shape_adt =
  | Square of point * float
  | Rect of point * float * float
  | Circle of point * float ;;

(*.....................................................................
Exercise 1A: Given the definitions above, write a function `area_adt`
that accepts a `shape_adt` and returns a `float` representing the area
of the shape. (You can review the area calculations for the various
shapes in Section B.4 of the textbook.)
.....................................................................*)

let area_adt (s : shape_adt) : float =
  match s with
  | Square (_, s) -> s *. s
  | Rect (_, w, h) -> w *. h
  | Circle (_, r) -> Float.pi *. r ** 2. ;;

(*.....................................................................
Exercise 1B: Write a function `list_area_adt` that, given a list of
elements of type `shape_adt`, returns a list of areas corresponding to
each shape.
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let list_area_adt : shape_adt list -> float list =
  List.map area_adt ;;

(*=====================================================================
Part 2: Interfaces, Classes, Objects

Why is implementing shapes as an ADT not ideal?

Suppose you travel to Flatland and meet a new shape, `Triangle`. What
must change above to support the new shape?

The type definition of shape needs to change to include `Triangle`:

type shape_adt =
  | ...
  | Triangle of point * point * point

and the area function (and more generally, any function that used a
match statement on `shape_adt`) would need to change to include the
`Triangle` case:

let area (s : shape_adt) : float =
  match s with
  | ...
```

```
  | Triangle ... -> ...

You've seen this problem before in the previous lab.

Thus, extending our world of shapes tends to break a lot of different
parts of the code before it starts to work again. In real production
code, we may not always have access to all elements (we may not have
access to the type definition, for example, or the area function), so
adding new functionality in this manner may be difficult or
impossible.

As you might imagine, declaring all possible shapes up front is an
equally poor idea.

Using algebraic data types gives us a *closed* definition for all
possible types in this world; this means that we must know all
possible variants at the time of type definition.

We can resolve this difficulty with object-oriented programming.

Below, we've created a class type (or interface). Interfaces define a
new type and specify methods for us to interact with objects of this
new type.

Once we have defined this class type, we can create new shapes by
defining classes that implement the shape interface. *)

class type shape =
  object
    (* The area of this shape *)
    method area : float

    (* The lower-left corner and the upper-right corner of the
       box that bounds the shape *)
    method bounding_box : point * point

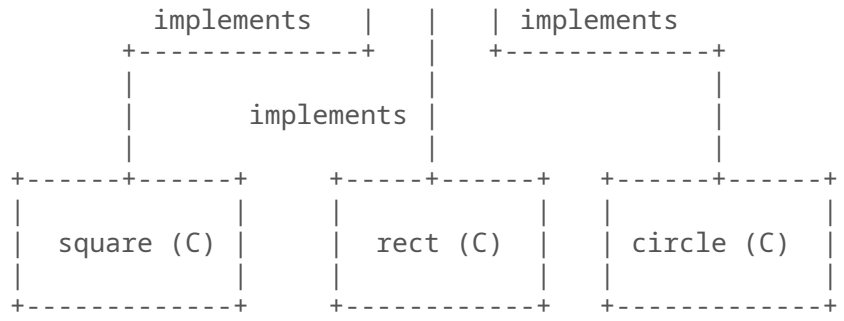
    (* The center point of this shape *)
    method center : point

    (* Translates the shape by the offset specified in the point *)
    method translate : point -> unit

    (* Dilates the shape by the scale factor *)
    method scale : float -> unit
  end ;;

(* This shape interface can have multiple classes implementing it, as so:

                         +-------------+
                         |             |
                         |  shape (I)  |
                         |             |
                         +--^---^---^-+
                            |   |   |
```



In the graphic above, an 'I' denotes a class type (interface) whereas a 'C' denotes a concrete implementation of that interface, a class. Concrete classes implement an interface class type, which are denoted by arrows and the label 'implements'.

Below, we'll create these classes that implement the `shape` class type.

A class is a specification for how to build objects; you might think of a class as a blueprint and an instance of the class as a specific building created with that blueprint. (A class type in this analogy might be a set of architectural regulations that blueprints need to satisfy.)

Classes may include:

  o Definitions of instance variables and methods. Each object, or instance, of a class has its own copy of instance variables.

  o Information on how to construct and initialize objects.

  o Scope information about what to hold private.

Here, the arguments to `rect` represent *constructor* arguments: values necessary to initialize the object.

Notice that the type of the `rect` class is `shape`, or more properly, the `rect` class implements the `shape` interface.

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Exercise 2A: Implement the `rect` class. To think about: How do we
store the values provided by the constructor? Keep in mind that the
`scale` and `translate` methods may need to modify aspects of the
object.
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class rect (initial_pos : point)
           (initial_width : float)
           (initial_height : float)
         : shape =
  object (this)

    (* instance variables that store the rectangle's properties *)
```

```
    val mutable pos = initial_pos  (* lower-left corner of rectangle *)
    val mutable width = initial_width
    val mutable height = initial_height

    method area : float =
      width *. height

    method bounding_box : point * point =
      let x, y = pos in
      pos, (x +. width, y +. height)

    method center : point =
      let (x1, y1), (x2, y2) = this#bounding_box in
      (x1 +. x2) /. 2., (y1 +. y2) /. 2.

    (* Destructively update `pos` to translate the shape by the values
       given by `vector`. *)
    method translate (vector_x, vector_y : point) : unit =
      let x, y = pos in
      pos <- x +. vector_x, y +. vector_y

  (* Scale the width and height of the rectangle from the lower-left
     corner by the scale factor `k`. *)
    method scale (k : float) : unit =
      width <- width *. k;
      height <- height *. k
  end ;;

(*.....................................................................
Exercise 2B: Implement the `circle` class.
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class circle (initial_center : point)
             (initial_radius : float)
           : shape =
  object
    val mutable center = initial_center
    val mutable radius = initial_radius

    method area : float = Float.pi *. radius ** 2.

    method bounding_box : point * point =
      let x, y = center in
      (x -. radius, y -. radius), (x +. radius, y +. radius)

    method center : point =
      center

    (* Destructively update position to translate the shape by the
       values given by `vector`. *)
    method translate ((vector_x, vector_y) : point) : unit =
      let x, y = center in
      center <- x +. vector_x, y +. vector_y

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    (* Scale the radius by scale factor `k` without moving its
       center. *)
    method scale (k : float) : unit =
      radius <- radius *. k

  end ;;

(*..........................................................................
Exercise 2C: Implement the `square` class. Notice how similar it is to
`rect`!
..........................................................................*)

class square (initial_pos : point) (initial_side : float) : shape =
  object (this)
    (* instance variables that store the square's properties *)
    val mutable pos = initial_pos  (* lower-left corner of square *)
    val mutable side = initial_side

    method area : float = side *. side

    method bounding_box : point * point =
      let x, y = pos in
      pos, (x +. side, y +. side)

    method center : point =
      let (x1, y1), (x2, y2) = this#bounding_box in
      (x1 +. x2) /. 2., (y1 +. y2) /. 2.

    (* Destructively update `pos` to translate the shape by the values
       given by `vector`. *)
    method translate ((vector_x, vector_y) : point) : unit =
      let x, y = pos in
      pos <- x +. vector_x, y +. vector_y

    (* Scale the sides of the square from the lower-left corner. *)
    method scale (k : float) : unit =
      side <- side *. k
  end ;;

(* Recall one of the original motivations for these exercises. We
wanted to create a single `area` function that returns the area of any
shape. Let's discover how easy this is with our objects.

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Exercise 2D: Create a function called `area` that accepts a shape
object and returns a `float` of the area for that shape. Hint: If your
definition isn't truly trivial, you're missing the point here.
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let area (s : shape) : float =
  s#area ;;

(*..........................................................................
Exercise 2E: Create a list of instantiated shapes called `s_list`.
The list should contain, in order:
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1. a `rect` at (1, 1) with width 4 and height 5
2. a `circle` at (0, -4) with radius 10
3. a `square` at (-3, -2.5) with size 4.3
.....................................................................*)

let s_list = [ new rect (1., 1.) 4.0 5.0;
               new circle (0., -4.) 10.;
               new square (-3., -2.5) 4.3 ] ;;

(* A DIGRESSION about a common confustication: As you might recall,
   lists can only contain objects of the same type. Why does the type
   system not show an error with your answer to 2E? What is the type
   of `s_list`?

   When you've completed this exercise, you might notice that the type
   reported for this list is `rect list`. Why is that, especially
   since not all the elements of the list are rectangles (!), and all
   elements of a list are supposed to be of the same type? The actual
   *type* associated with the elements of the list is an "object
   type", as described in Real World OCaml
   <https://dev.realworldocaml.org/objects.html>, in particular,
   something like:

    < area : float;
      bounding_box : point * point;
      center : point;
      scale : float -> unit;
      translate : point -> unit >

   The `rect`, `circle`, and `square` objects are *all* of this
   type. But the ocaml REPL tries to be helpful in printing a more
   evocative name for the type. By treating the class names as
   synonyms for the object types, the type of the list can be
   interpreted as a `rect list` or `circle list` or `square list`. The
   REPL uses the class of the first element in the list as the type
   name to use, thus `rect list`. Had the elements been in another
   order, the type might have been abbreviated as `circle list` or
   `square list`. END DIGRESSION *)

(*======================================================================
Part 3: Representation, Inheritance

You might have noticed that `rect` and `square` are very similar
representations, or implementations, of a class. They both use a
`point` to represent a lower-left corner, and both take side
length(s). A square, as you know, is a rectangle with the additional
constraint that its width is equal to its height.

We can reimplement `square` to *inherit* from `rect`, and thus rely on
`rect`'s existing representation.

In the end, we'll have this revised type hierarchy:

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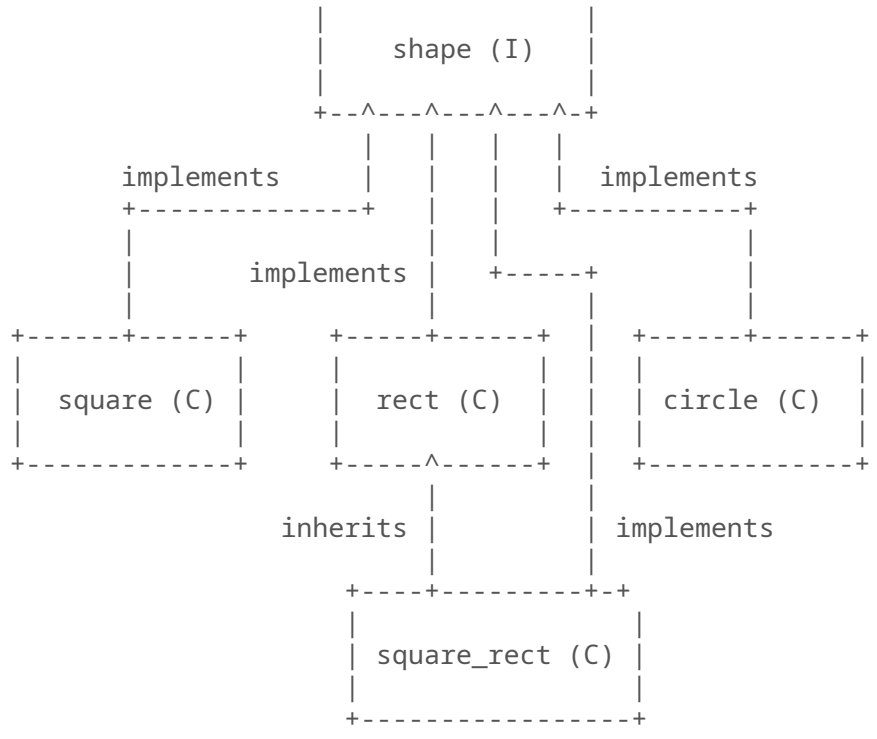


```
(*.....................................................................
Exercise 3A: Implement the `square_rect` class which inherits all of
its methods from its parent.

NOTE:

    In this and some later problems, we comment out the original
    stub code because it won't even compile yet until you finish the
    problem. Consequently, your code won't compile on Gradescope until
    you complete these problems.
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class square_rect (pos : point) (side : float) : shape =
  object
    inherit rect pos side side
  end ;;

(*.....................................................................
Exercise 3B: Now, implement a `square_center_scale` class that
inherits from `square_rect`, but *overrides* the `scale` method so
that the center (rather than the lower-left corner) of the square
stays in the same place.

You may find this tricky because *you don't have access to the
instance variables of the class that you inherit from*. (Why is this?)
A hint: First scale, then translate the center back to its original
position.
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class square_center_scale (pos : point) (side : float) : shape =
```

```
  object (this)
    inherit square_rect pos side as super

    method! scale (k : float) : unit =
      let x1, y1 = this#center in
      (* scale *)
      let _ = super#scale k in
      let x2, y2 = this#center in
      (* translate back to center *)
      this#translate (x1 -. x2, y1 -. y2)
  end ;;

(* Some in lab were enticed to try to directly refer to the `pos`
   instance variable inherited from the `square_rect` class. But that
   instance variable is not available to be referred to -- either
   implicitly as `pos` or explicitly as `this#pos` or `super#pos` --
   because the class type of the superclass `square_rect`, `shape`,
   does not reveal the `pos` instance variable. Only the five methods
   specified in the shape class type are available to use.

   Were we not to specify that the class satisfy the `shape` class
   type, it would have been possible to refer directly to the
   inherited `pos` instance variable. Though this might allow some
   brevity, it would do so at the cost of eliminating the important
   information-hiding role of the class type, introducing a design
   flaw in the overall system; classes should generally be explicit
   about their signatures (via class types) to provide a strong
   abstraction barrier.

   Why refer to `this#center` versus `super#center`? By virtue of
   inheritance, these refer to the same method, so either will give
   the same result. Nonetheless, we might prefer `this#center` in
   order to "future-proof" the code in case the `center` or
   `translate` methods are overridden later. *)

(* Before we move on, consider: do you need to make any modifications
to the `area` function you wrote in Exercise 2D to support these new
classes? *)

(* ANSWER: No, you do not! *)

(*======================================================================
Part 4: Subtyping Polymorphism and Dynamic Dispatch

As we wander more around Flatland, we discover that there are more
four-sided shapes than we originally thought. We knew about Square and
Rect, but we've also seen Rhombi, Trapezoids, and other four-sided
creatures that are collectively called Quadrilaterals.

Since Square and Rect both like to identify themselves as
Quadrilaterals, which also identify themselves as Shapes, we need to
make Quadrilateral a *subtype* of Shape.

Below, we have defined a new class type (interface) called
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```
`quad`. Notice that `quad` has all of the methods in `shape`'s
signature, but adds an additional method, `sides`, that returns the
lengths of each of the four sides.

Since `quad` can do everything that a shape can do (and thus, wherever
we expect a `shape`, we can safely pass a `quad`), we consider `quad`
a *subtype* of `shape`. *)

class type quad =
  object
    inherit shape

    (* returns the lengths of the four sides *)
    method sides : float * float * float * float
  end ;;

(* We can revise the type hierarchy as below, adding the `quad` class
type and some quadrilateral classes `square_quad` and `rect_quad`,
allowing us to drop `square` and `rect` but keeping `circle`.

                  +------------+
                  |            |
                  |  shape (I) |
                  |            |
                  +--^---^-----+
                     |   |
        implements   |   |
            +--------+   +-----+
            |                  |
            |         subtypes |
            |                  |
     +------+------+     +-----+------+
     |             |     |            |
     |  circle (C) |     |  quad  (I) |
     |             |     |            |
     +-------------+     +-^----^-----+
                         |    |
              implements |    | implements
            +------------+    +--------+
            |                         |
            |                         |
    +-------+---------+             +----+-----------+
    |                 |  inherits   |                |
    | square_quad (C) +-------------> rect_quad (C) |
    |                 |             |                |
    +-----------------+             +----------------+


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Exercise 4A: Write the class `rect_quad`, which represents a rectangle
that implements a `quad` class type. Hint: By taking advantage of
existing classes, you should only need to implement a single method.
.....................................................................*)

class rect_quad (initial_pos : point)
```

```
                (initial_width : float)
                (initial_height : float)
              : quad =
  object
    inherit rect initial_pos initial_width initial_height as super

    method sides : float * float * float * float =
      let (x1, y1), (x2, y2) = super#bounding_box in
      let width = x2 -. x1 in
      let height = y2 -. y1 in
      width, height, width, height
  end ;;

(* We can't just use `initial_width` and `initial_height` in the
   return value, since the actual width and height may have changed
   since. Instead, we make use of the `bounding_box` method to reveal
   the current width and height of the rectangle.

   Note that we've referred to `super#bounding_box` here. But because
   of late binding, we could just as well have referred to
   `this#bounding_box`, which ends up referring to the same method. *)

(*.....................................................................
Exercise 4B: Complete a class `square_quad` that represents a square
that implements a `quad` class type. Hint: you shouldn't need to
implement any methods!
.....................................................................*)

class square_quad (p : point) (s : float) : quad =
  object
    inherit rect_quad p s s
  end ;;

(* Remember Exercise 2D, in which you implemented an area function for
shapes? Amazingly, even though we have continued to create new shapes,
due to subtyping and inheritance we can still rely on the existing
implementation you already created! *)

(*.....................................................................
Exercise 4C: Create an instance of `square_quad` and name it
`sq`. Then, pass it to the area function to find out its area and
store the result in a variable `sq_area`.

Hint: If you find yourself having problems with what ought to be a
simple exercise, see the discussion of subtype coercion in Section
18.5.
.....................................................................*)

let sq : quad = new square_quad (3., 4.) 5. ;;

let sq_area = area (sq :> shape) ;;

(* The natural inclination is to write
```

```
      let sq_area = area sq ;;

   But this fails to type, since OCaml does not perform type inference
   for object subtypes. Instead, we must be explicit about the subtype
   to supertype (`quad` to `shape`) coercion by using the `:>`
   operator. *)

(*.....................................................................
Exercise 4D: Write a function `area_list` that accepts a list of
shapes and returns a list of their areas.
.....................................................................*)

let area_list : shape list -> float list =
  List.map area ;;

(* This works because of *dynamic dispatch*; we decide the code to run
   at run-time instead of compile-time. In other words, the
   `shape#area` call in the `area` function determines at run-time the
   specific method to call based on the object passed to it.

   Compare this to the `area_adt` function, which is not dynamic
   because the same code is run every time. Even though the match case
   may not be known, the branch is wholly contained within that static
   function. *)
```